

WEEK 09 REPORT – ENTERPRISE NETWORKING

1. Introduction

Overview of Enterprise Networking

Enterprise networking refers to the design, implementation, and management of large-scale computer networks used by organizations to support business operations, communication, data sharing, and security. Unlike small home networks, enterprise networks require structured configurations, segmentation, scalability, and fault tolerance to handle multiple users, departments, and services efficiently.

In enterprise environments, technologies such as **VLANs (Virtual Local Area Networks)**, **Inter-VLAN Routing**, **trunking**, and **routing protocols** are widely used to improve network performance, enhance security, and simplify network management. These technologies help isolate traffic, reduce broadcast domains, and enable controlled communication between different network segments.

Purpose of Week 9 Labs

The purpose of Week 9 labs was to gain **hands-on experience** with enterprise-level networking concepts using **Cisco Packet Tracer**. The labs focused on understanding how logical network segmentation works and how routers and switches interact in a multi-VLAN environment.

By performing these labs, the objective was to:

- Understand VLAN creation and management
 - Implement Inter-VLAN routing using routers
 - Configure trunk and access ports
 - Troubleshoot real-world networking issues
 - Build confidence in enterprise network design and debugging
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2. What I Learned

VLAN and Inter-VLAN Routing

One of the most important concepts learned during this week was **VLAN configuration**. VLANs allow a single physical switch to be divided into multiple logical networks. Devices within the same VLAN can communicate directly, while communication between different VLANs requires routing.

I learned how to:

- Create VLANs on switches
- Assign switch ports to specific VLANs
- Understand the importance of VLAN IDs and consistency

Inter-VLAN routing was implemented to allow communication between different VLANs. This demonstrated how routers act as a gateway between logically separated networks.

Router Sub-Interfaces

Router sub-interfaces were used to enable Inter-VLAN routing using a single physical router interface. Each sub-interface was assigned:

- A VLAN ID using encapsulation dot1Q
- A unique IP address acting as the default gateway for that VLAN

This concept helped me understand how a single router interface can handle traffic from multiple VLANs using VLAN tagging.

Trunking

Trunking was another critical concept learned during this week. A trunk port allows multiple VLANs to pass through a single physical link between networking devices, such as a switch and a router or between two switches.

Key learning points:

- Difference between **access ports** and **trunk ports**
 - Importance of enabling trunking on correct interfaces
 - How incorrect trunk configuration can completely block Inter-VLAN communication
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Routing Protocols Basics

Although the primary focus was VLAN-based routing, basic knowledge of routing concepts was reinforced. I gained clarity on:

- Difference between Layer 2 switching and Layer 3 routing
- Role of routers in forwarding packets between networks
- Importance of IP addressing and default gateways

This foundation is essential for understanding advanced routing protocols in future labs.

3. Screenshots (Attached in GitHub Folder)

The following screenshots were captured and included as part of the Week 9 documentation:

- **Network topology diagrams** showing routers, switches, and connected PCs
- **VLAN configuration outputs**
- **Router interface and sub-interface configurations**
- **Successful ping outputs** demonstrating Inter-VLAN communication

These screenshots serve as proof of correct configuration and successful implementation of enterprise networking concepts.

4. Problems Faced & Solutions

During the lab exercises, several issues were encountered, which helped develop troubleshooting skills. The major problems and their solutions are summarized below:

Problem Faced	Solution Applied
Ping failed between VLANs	Verified and corrected default gateway on PCs
VLAN mismatch	Reassigned switch ports to correct VLAN IDs
Incorrect VLAN created	Deleted wrong VLAN and recreated correct VLAN
No trunk between router and switch	Enabled trunk mode on the correct interface
Interface down	Used no shutdown to activate interfaces
Routing failure	Corrected router interface IP addressing

These issues highlighted the importance of **step-by-step verification**, attention to detail, and logical troubleshooting when working with enterprise networks.

5. Conclusion

Week 9 provided valuable hands-on experience in **enterprise-level networking**. Through practical labs, I gained a strong understanding of VLANs, Inter-VLAN routing, trunking, and router–switch interaction. The challenges faced during configuration helped improve my troubleshooting and analytical skills.

This week significantly enhanced my confidence in designing and managing structured networks and laid a strong foundation for more advanced networking and security concepts in future weeks. Overall, the labs successfully bridged the gap between theoretical knowledge and real-world network implementation.
